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Centre Number

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Student Number

SCEGGS Darlinghurst

2006

**Preliminary Course
Semester 2 Examination**

Mathematics

General Instructions

- Reading time – 5 minutes
- Working time – 2 hours
- Write using blue or black pen
- Write your Centre Number and Student Number at the top of each page
- Attempt **all** questions and show all necessary working
- Marks will be deducted for careless or badly arranged work
- Mathematical templates, geometrical equipment and scientific calculators may be used
- **Start each question on a new page**
- **Do not** attach all question together in one bundle

Total marks – 76

- Attempt Questions 1–6

Question	Comm	Calc	Reason	Total
1	/2		/2	/13
2				/12
3	/1	/4	/6	/13
4	/2		/2	/14
5		/3	/6	/12
6	/6	/2		/11
TOTAL	/11	/9	/16	/76

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Total marks – 76

Attempt Question 1 – 6

Attempt **all** questions on the pad paper provided

Write your Centre Number and Student Number at the top of each page

Show all necessary working

Marks may be deducted for careless or badly arranged work

Mathematical templates, geometrical equipment and scientific calculators may be used

Start each question on a new page

	Marks
Question 1 (13 marks)	
(a) Evaluate correct to 2 decimal places.	1
$\frac{3.65 - 2.3^2}{6.4 - 8.39}$	
(b) Simplify fully:	
$\sqrt{72} + 2\sqrt{18}$	2
(c) Solve: $3x^2 - 15 = 0$	2
(d) Express with a rational denominator in simplest form.	2
$\frac{\sqrt{3} + 1}{\sqrt{3} - 1}$	
(e) Solve: $27^{2x} = 9$	2
(f) Factorise fully: $8x^3 + 125$	2
(g) Solve $ x + 5 < 3$ and graph your solution on a number line	2

Question 2 (12 marks)

(a) Given $f(x) = x - \frac{1}{x}$

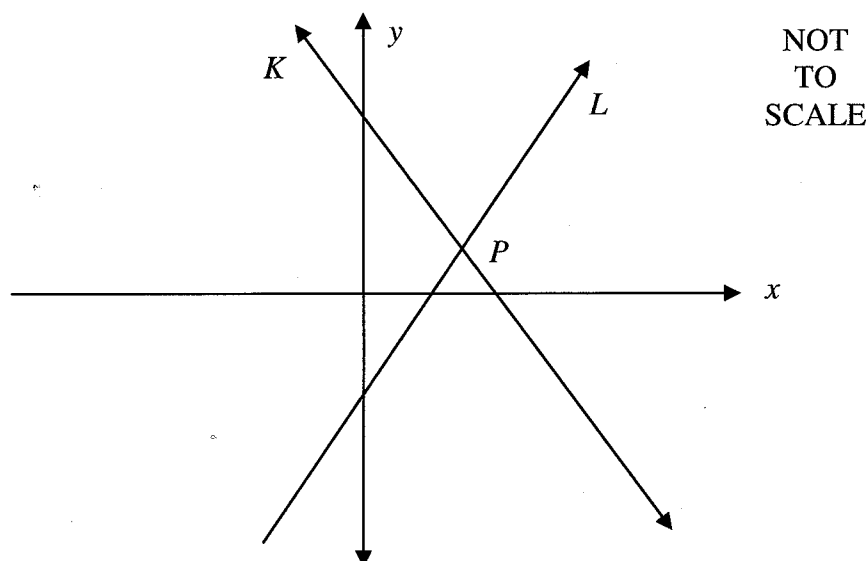
(i) find $f(-2)$ and $f\left(\frac{1}{2}\right)$

2

(ii) if $f(a) = 0$ find a

1

(b)



(i) Two straight lines L and K with equations $3x - 2y = 4$ and $y = -2x + 5$ respectively, meet at the point P as shown in the diagram above. Find the co-ordinates of P .

2

(ii) Write down the gradient and y-intercept of the line L .

2

(iii) Calculate the area of the triangle formed by the lines L , K and $x = 0$.

1

(iv) Find the equation of the line J which is parallel to L and which passes through the point $(0, 5)$.

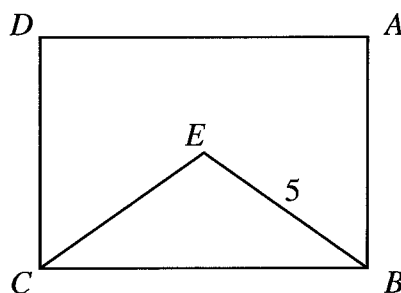
2

(v) Calculate the exact perpendicular distance between the parallel lines L and J .

2

Question 3 (13 marks)

- (a) The interior angle of a regular polygon is 170° . Find the number of sides. 2
- (b) Simplify $(\tan \theta + \sec \theta)(\sec \theta - \tan \theta)$ 2
- (c) Differentiate the following:
- (i) $y = 3x - \frac{5}{x^2}$ 2
- (ii) $y = \frac{x^3 - 1}{x + 1}$ 2
- (d) In the diagram below $ABCD$ is a square and EC and EB are the bisectors of $\angle DCB$ and $\angle ABC$ respectively.



NOT
TO
SCALE

- (i) Copy the diagram onto your answer sheet and show all given information. 1
- (ii) Prove that $\triangle BCE$ is isosceles. 2
- (iii) Hence find the exact length of BC . 2

Question 4 (14 marks)

- (a) Find the quadratic equation whose roots are 1 and -4 . 1
- (b) If α and β are the roots of the quadratic equation $x^2 - 5x + 2 = 0$, find:
- (i) $\alpha + \beta$ 1
- (ii) $\alpha\beta$ 1
- (iii) $\frac{1}{\alpha^2} + \frac{1}{\beta^2}$ 2
- (c) Given $\sin \theta = \frac{3}{5}$ and $\tan \theta < 0$, find $\sec \theta$. 2
- (d) Consider the function $y = \frac{2}{x+1}$
- (i) State the Domain and Range of the function. 2
- (ii) Is this function continuous? Clearly explain your answer. 1

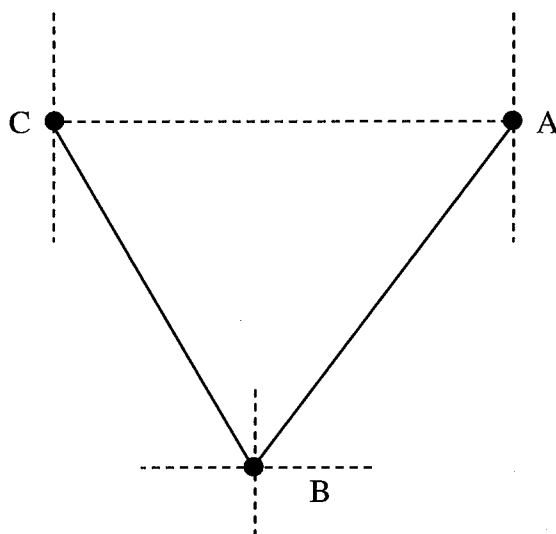
Question 4 continues on the next page

Question 4 (continued)

- (e) A yacht sails from a port A on a course bearing of 212° for 2.8 nautical miles to B . It then turns and sails on a course bearing of 330° to a point C , due west of A .

- (i) Copy the diagram below onto your answer sheet and show all given information.

1



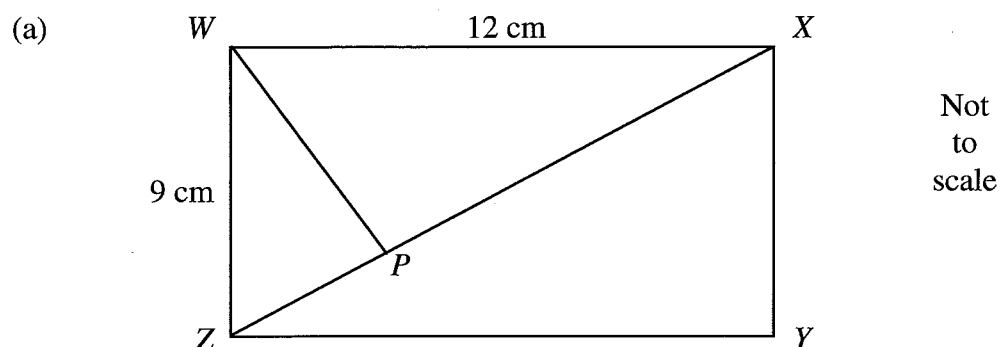
- (ii) Find $\angle ACB$.

1

- (iii) Hence, find to the nearest tenth of a nautical mile, the distance BC .

2

Question 5 (12 marks)



In the diagram above, $WXYZ$ is a rectangle with $WX = 12$ cm and $WZ = 9$ cm. WP is perpendicular to XZ .

Copy the diagram onto your answer sheet.

- | | | |
|-------|---|---|
| (i) | Find the length of XZ . | 1 |
| (ii) | Prove that $\triangle WXP$ is similar to $\triangle ZXW$. | 3 |
| (iii) | Hence find the length of XP , give reason(s). | 2 |
| (b) | Solve $4\cos^2 \theta = 3$ if $0^\circ \leq \theta \leq 360^\circ$. | 3 |
| (c) | Find the equation of the normal to the curve $y = (2x - 5)^4$ at $x = 2$ in general form. | 3 |

Question 6 (11 marks)

(a) Find the values of p for which the expression $3x^2 + px + 3$ is positive definite. **3**

(b) On a number plane shade the region(s) which represent the simultaneous solution of: **4**

$$y < x^2 \text{ and } y \geq |x - 1|$$

(c) (i) Explain why the gradient of $y = \frac{1}{x}$ is always negative. **2**

(ii) Find the gradient of l , the tangent of $y = \frac{1}{x}$ at $x = \frac{1}{2}$. **1**

(iii) What angle to the nearest degree, does the line l make with the positive direction of the x -axis. **1**

End of paper

Yr 11 Preliminary Mathematics Semester 2 Exam 2006

Question 1 (13 Marks) MF

a) 0.82 ✓

Put a bracket around the denominator before dividing if you got this one wrong.

b) $6\sqrt{2} + 6\sqrt{2} = 12\sqrt{2}$ ✓

Note $2\sqrt{8} = 2 \times 3\sqrt{2} = 6\sqrt{2}$
↑ multiply don't add!

c) $3x^2 = 15$
 $x^2 = 5$ ✓
 $x = \pm\sqrt{5}$ ✓

If you solve this way using difference of 2 squares you get the same answers.

$$3x^2 - 15 = 0$$

$$3(x^2 - 5) = 0$$

$$3(x - \sqrt{5})(x + \sqrt{5}) = 0$$

↑ $x = \sqrt{5}, x = -\sqrt{5}$

Note: This 3 is only a numerical factor
It does not give a solution of 0.

d) $\frac{\sqrt{3}+1}{\sqrt{3}-1} \times \frac{\sqrt{3}+1}{\sqrt{3}+1}$
 $= \frac{3 + \sqrt{3} + \sqrt{3} + 1}{3 - 1}$
 $= \frac{4 + 2\sqrt{3}}{2}$ ✓
 $= 2 + \sqrt{3}$ ✓

An easy question!!

This part was not very well done.

More practice at expanding binomial products is recommended if you got this one wrong.

← Simplest form required.

e) $(3^3)^{2x} = 3^2$
 $3^{6x} = 3^2$ ✓
 $6x = 2$
 $x = \frac{1}{3}$ ✓

(R
2)

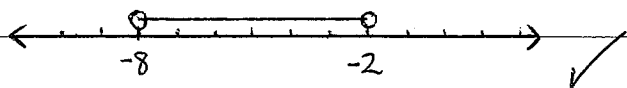
* Poorly done. Index Rules are often used to solve exponential equations.

f) $(2x+5)(4x^2-10x+25)$

Learn the rule $a^3 + b^3 = (a+b)(a^2 - ab + b^2)$ and how to use it.

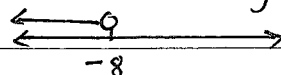
g) $x+5 < 3$ $-(x+5) < 3$
 $x < -2$ $x+5 > -3$ ✓ both solutions
 $x > -8$

• First mark awarded for solving two cases correctly.



(C
2)

If you get the solution $x < -8, x < -2$ your graph should show only $x < -8$.



Question 2 (12 Marks) AV

xi) $f(-2) = -2 - \frac{1}{-2}$
 $= -1\frac{1}{2}$ ✓

Generally well done, be careful with your negatives

$f(\frac{1}{2}) = \frac{1}{2} - \frac{1}{\frac{1}{2}}$
 $= \frac{1}{2} - 2$
 $= -1\frac{1}{2}$ ✓

* If you don't know $\frac{1}{2} - 2 = -1\frac{1}{2}$ then use your calculator!!

ii) $0 = a - \frac{1}{a}$
 $\frac{1}{a} = a$
 $1 = a^2$
 $a = \pm 1$ ✓

Not well done

$f(a) = 0$ does not mean substitute 0 for a

bi) L: $3x - 2y = 4$ K: $y = -2x + 5$

Generally well done

P: $3x - 2(-2x + 5) = 4$
 $3x + 4x - 10 = 4$
 $7x = 14$
 $x = 2$ ✓

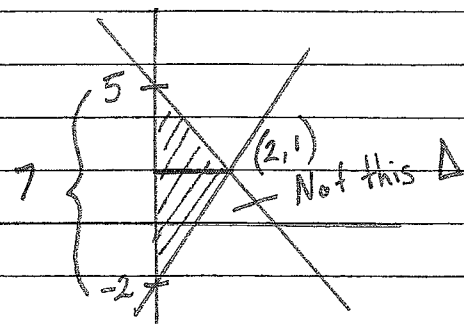
From K

$y = -2 \times 2 + 5$
 $= 1$

$\therefore P(2, 1)$ ✓

ii) $3x - 2y = 4$
 $-2y = 4 - 3x$
 $y = -2 + \frac{3x}{2}$
 $m = \frac{3}{2}$ ✓ $b = -2$ ✓

"



iii) $A = \frac{1}{2} \times 7 \times 2$
 $= 7$ ✓

Poorly done - Most did not read the question carefully and tried to find the area of the wrong Δ
 * If a question is worth 1 mark it is not going to involve a page of working

iv) parallel to $L \therefore m = \frac{3}{2}$

through $(0, 5)$

$$\text{Eqn } j: y - 5 = \frac{3}{2}(x - 0)$$

$$2y - 10 = 3x \quad \checkmark$$

$$3x - 2y + 10 = 0$$

Not perpendicular - read the question

Generally well done

v) $L: 3x - 2y - 4 = 0$

through $(0, 5)$

$$d = \frac{|3 \times 0 + -2 \times 5 + -4|}{\sqrt{3^2 + (-2)^2}} \quad \checkmark$$

$$d = \frac{14}{\sqrt{13}} \times \frac{\sqrt{13}}{\sqrt{13}}$$

$$d = \frac{14\sqrt{13}}{13} \quad \checkmark$$

OR

$$J: 3x - 2y + 10 = 0$$

through $P(2, 1)$

$$\begin{aligned} d &= \frac{|3 \times 2 + -2 \times 1 + 10|}{\sqrt{3^2 + (-2)^2}} \\ &= \frac{|14|}{\sqrt{13}} \end{aligned}$$

OK.

Make sure you choose a point that is not on the line

- quite a few did this and ended up with a distance of 0

- if this happens check what line and point you have chosen.

Question 3 (13 Marks) TS

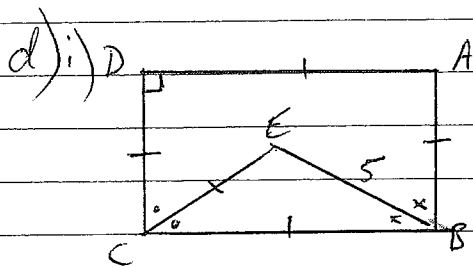
a) Ext $\angle = 10^\circ$ ✓
 $n = 360 \div 10$
 $n = 36$ ✓

b) $\sec^2 \theta - \tan^2 \theta$
 $= 1 + \tan^2 \theta - \tan^2 \theta$ ✓
 $= 1$ ✓ (R 2)

c) i) $y = 3x - 5x^{-2}$
 $y' = 3 + 10x^{-3}$ ✓✓

ii) $u = x^3 - 1$ $v = x + 1$
 $u' = 3x^2$ $v' = 1$

(Ca 4) $y' = \frac{3x^2(x+1) - (x^3-1)}{(x+1)^2}$ ✓
 $= \frac{3x^3 + 3x^2 - x^3 + 1}{(x+1)^2}$
 $= \frac{2x^3 + 3x^2 + 1}{(x+1)^2}$ ✓



✓ must show equal angles and square
 (C 1)

ii) $\angle ABE = \angle CBE$ (EB bisects $\angle ABC$)
 $= 45^\circ$ ($\angle ABC = 90^\circ$, vertex of a square)
 $\angle DCE = \angle BCE$ (EC bisects $\angle DCB$)
 $= 45^\circ$ ($\angle DCB = 90^\circ$, vertex of a square)
 $\therefore \angle ECB = \angle EBC$
 $\therefore \triangle BCE$ is isosceles
 (2 equal angles)

Note:
 170° is NOT the whole $\angle$ sum!

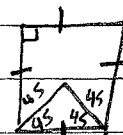
or $180(n-2) = 170n$ ✓
 $n = 36$ ✓

$\frac{5}{x^2}$ does NOT need the quotient rule to be differentiated

Lots of people had the correct answer & then ruined it by bad algebra!

is NOT a square!
 It is only a rhombus.

✓ This was very badly done.
 And when you are told to mark the information given that is exactly what you need to do — do not deduce anything



Is not correct!

ii) $\angle CEB = 90^\circ$ (L sum $\triangle CEB = 180^\circ$) ✓
 $CE = EB = 5$ (sides opp = $\angle$ s of isos $\triangle$ =)
 $BC^2 = 5^2 + 5^2$
 $BC^2 = 50$
 $BC = 5\sqrt{2}$ ✓

(R
2)

MUST give reasons here

- why is $\angle CEB = 90^\circ$?

- why does $CE = EB = 5$?

It is NOT given!

Question 4 (14 Marks) SL

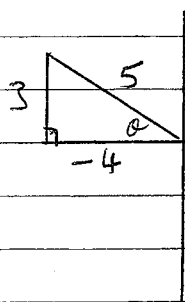
i) $(x-1)(x+4) = 0$
 $x^2 - x + 4x - 4 = 0$
 $x^2 + 3x - 4 = 0$ ✓

Please remember a quadratic equation
EQUALS ZERO !!!

bi) $\alpha + \beta = 5$ ✓
 ii) $\alpha\beta = 2$ ✓
 iii) $\frac{\beta^2 + \alpha^2}{(\alpha\beta)^2} = \frac{(\alpha + \beta)^2 - 2\alpha\beta}{(\alpha\beta)^2}$ ✓
 $= \frac{5^2 - 2 \times 2}{4}$
 $= \frac{21}{4}$ ✓

$$\alpha^2 + \beta^2 \neq (\alpha + \beta)^2$$

c) sin +ve, tan -ve in Q2



$$\cos \theta = \frac{-4}{5}$$

$$\sec \theta = \frac{5}{-4}$$

(R
2)

Be careful with quadrants

2nd quadrant means $\cos \theta$ is negative

$\therefore \sec \theta$ is negative

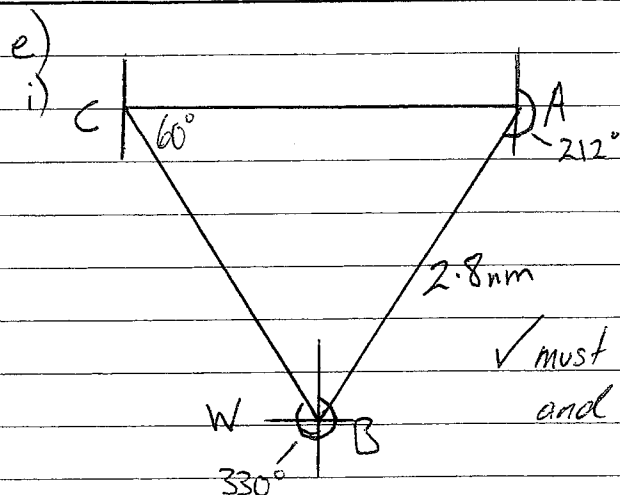
d) i) D: all real x , $x \neq -1$ ✓
 R: all real y , $y \neq 0$ ✓

You must say all real x
 & real y .

ii) Not continuous as there is
 a vertical asymptote at
 $x = -1$

Need to give mathematical reasons

✓ (C)
 (1)



Done well ✓✓

✓ (C)
 (1)

✓ must show bearing at A, B
 and length of AB

ii) $\angle CBW = 330^\circ - 270^\circ$
 $= 60^\circ$
 $\angle ACB = \angle CBW$ (alt. $\angle$ s = on // lines)
 $= 60^\circ$ ✓

iii) $\angle CAB = 270^\circ - 212^\circ$
 $= 58^\circ$ ✓

$$\frac{BC}{\sin 58^\circ} = \frac{2.8}{\sin 60^\circ}$$

$$BC = 2.7 \text{ nm}$$
 ✓

Question 5 (12 Marks) MF

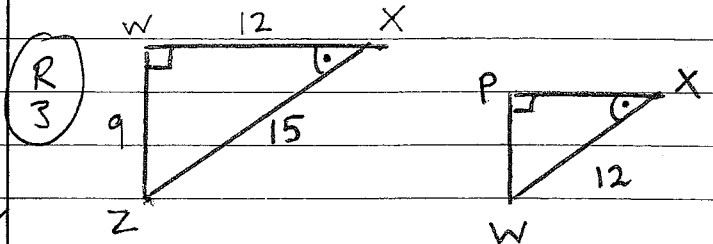
i) $XZ^2 = 12^2 + 9^2$ (By Pythagoras)
 $XZ^2 = 225$
 $XZ = 15$ ✓

ii) $\angle WXP$ is common ✓
 $\angle ZWX = 90^\circ$ (vertex of a rectangle) ✓
 $\angle WPX = 90^\circ$ ($WP \perp XZ$) ✓

$\therefore \triangle WXP \parallel \triangle ZWX$ (equiangular) ✓

iii) $\frac{XP}{12} = \frac{12}{15}$ (corr. sides of sim Δ s in equal ratio) ✓
 $XP = \frac{144}{15}$
 $XP = 9.6 \text{ cm}$ ✓

Note that $\triangle XWZ$ is right-angled so Pythagoras is the most efficient method. Using Trig. is more difficult especially for only one mark.



Draw the triangles carefully in the same orientation. Then it's easy!

- Make sure you use correct sides!
- You must put a reason for the second mark.

b) $4 \cos^2 \theta = 3$
 $\cos^2 \theta = \frac{3}{4}$
 $\cos \theta = \pm \frac{\sqrt{3}}{2}$ ✓

(R
3)

$\therefore$ soln is $\theta = 1, 2, 3, 4$
acute $\angle = 30^\circ$
 $\therefore \theta = 30^\circ, 150^\circ, 210^\circ, 330^\circ$ ✓✓

- First mark for knowing there are two cases $\pm$
- Second mark for solving correctly $\theta = 30^\circ \& 330^\circ$.

(*) This is a standard question. It was poorly done.

c) $y = (2x-5)^4$
 $y' = 8(2x-5)^3$ ✓ at $x=2$ $y=1$
 $m_t = 8(4-5)^3$
 $m_t = -8$ $\therefore m_N = \frac{1}{8}$ ✓

(Ca
3)

Egn: $y-1 = \frac{1}{8}(x-2)$
 $8y-8 = x-2$
 $x-8y+6=0$ ✓ general form

Note $(-1)^3 = -1$
many mistakes made with this negative sign.

This is an easy calculus question. Practise more of the same type if you got it wrong.

Question 6 (11 marks) AV

a) pos def. when $a > 0$
 $\Delta < 0$

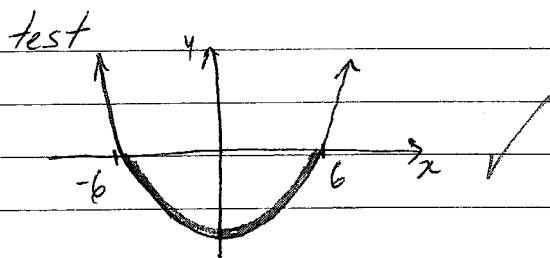
$$a = 3 \therefore a > 0$$

$$\Delta < 0$$

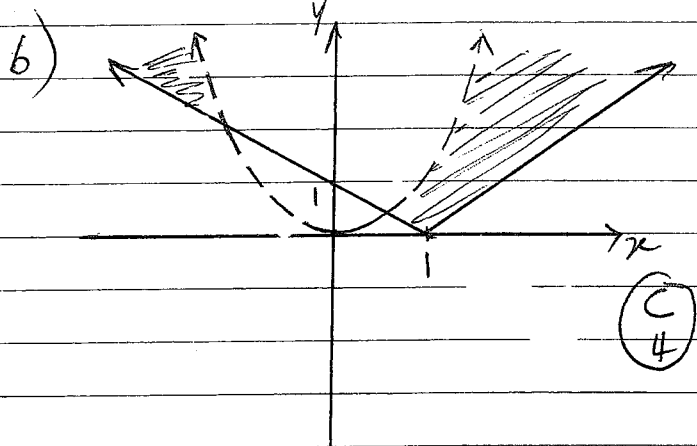
$$p^2 - 4 \times 3 \times 3 < 0$$

$$p^2 - 36 < 0$$

$$p^2 < 36 \quad \checkmark$$



$$\therefore -6 < p < 6 \quad \checkmark$$



Not well done at all
 most new to solve for $\Delta < 0$
 and got down to

$$p^2 - 36 < 0$$

To solve from here you
MUST sketch the
 quadratic inequality

* If you can't sketch get a Mathaid !!!
 $\checkmark y = x^2$
 $\checkmark y = |x-1|$ } lines most could sketch $y = x^2$
 $\checkmark$ both regions $y = |x-1|$ was not
 well done

Make sure you sketch to a reasonable
 scale so that the two regions can
 be easily shown.

* label axes & x & y intercepts.

c i) $y = x^{-1}$ y' represents the gradient
 $y' = -x^{-2}$ function of y , since
 $= -\frac{1}{x^2}$ x^2 must be positive
 $-\frac{1}{x^2}$ must therefore
 be negative $\checkmark$

(OR sketch showing gradient
 is always negative)

Extremely poorly done
 you must explain why a gradient
 function of $-\frac{1}{x^2}$ is always
 negative (ie x^2 is always +ve)
 $\therefore -\frac{1}{x^2}$ is always -ve

detailed
 If you did a clearly diagram showing
 that the gradient is -ve at any
 point by giving examples this was
 also acceptable.

ii) m_t at $x = \frac{1}{2}$

$$y' = -\frac{1}{x^2}$$

$$m_t = -\frac{1}{\left(\frac{1}{2}\right)^2}$$

$$= -1 \div \frac{1}{4} \quad \checkmark$$

$$= -4$$

Ca
1

Ok, not great.

you must differentiate then
substitute $x = \frac{1}{2}$

iii) $\tan Q = m$

$$\tan Q = -4$$

$\tan$ -ve in Q2

$$Q = 180 - 76^\circ$$

$$Q = 104^\circ \quad \checkmark$$

Ca
1

Not too bad

Remember if the gradient is
negative the angle will be
obtuse.